

In-orbit test of the weak equivalence principle with atom interferometry

DAN-FANG ZHANG , JING-TING LI , [...], AND MING-SHENG ZHAN  +22 authors [Authors Info & Affiliations](#)

SCIENCE ADVANCES 28 Aug 2026 Vol 12, Issue 35 DOI: 10.1126/sciadv.aeh4502

889



Abstract

The weak equivalence principle (WEP) is a central pillar of general relativity. Its precise test with quantum systems in space offers a unique window onto new physics. Here, we report an in-orbit quantum test of the WEP. A dual-species ($^{85}\text{Rb}/^{87}\text{Rb}$) atom interferometer is realized aboard the China Space Station. Methods of platform motion suppression, fluorescence detection switching, and two-photon detuning switching are developed to eliminate phase noise and improve measurement accuracy. A test uncertainty of 2.8×10^{-8} is obtained from 280 days of WEP test data, and a test result of $(-2.7 \pm 4.7) \times 10^{-7}$ is achieved after error estimation. This improves prior atom-interferometric WEP tests in microgravity by three orders of magnitude. This work paves the way for space-borne quantum inertial sensors and their application to future fundamental physics in space.

SIGN UP FOR THE AWARD-WINNING SCIENCEADVISER NEWSLETTER

The latest news, commentary, and research, free to your inbox daily

SIGN UP >

INTRODUCTION

The weak equivalence principle (WEP), which posits the universality of free fall, is a cornerstone of Einstein’s theory of general relativity. Its continued validation through ever more precise experiments is essential, as any detected violation would herald new physics (1). Tests using macroscopic bodies have reached phenomenal precision, at the 10^{-13} level on Earth (2, 3) and 10^{-15} in space (4, 5).

abling comparisons of atoms in different internal quantum states, such as those differing in spin (22, 26), superposition (27), or hyperfine structure (30, 32).

The precision of CAI scales quadratically with the interference time T . On Earth, the interference time is limited to a few seconds due to gravitational acceleration. Microgravity environments remove this constraint, enabling extended interference times. Pioneering experiments on microgravity platforms, including drop towers (34, 35), parabolic flight aircraft (25, 36), sounding rockets (37, 38), and the Einstein Elevator (39–41), have demonstrated the potential of this approach, with the first microgravity CAI-based WEP test reaching a precision of 10^{-4} (25). However, these platforms offer microgravity durations on the order of seconds or minutes. Reaching an extremely high WEP test precision, e.g., 10^{-17} (42, 43), requires a permanent microgravity environment for long-term data accumulation and comprehensive control of systematic errors. An orbital laboratory, such as the China Space Station (CSS) (44, 45) or the International Space Station (46–48), provides the necessary stable environment to realize this potential. Here, we report on the realization of a dual-species atom interferometer on board the CSS (Fig. 1A) and present the first in-orbit test of the WEP using CAI.

PDF

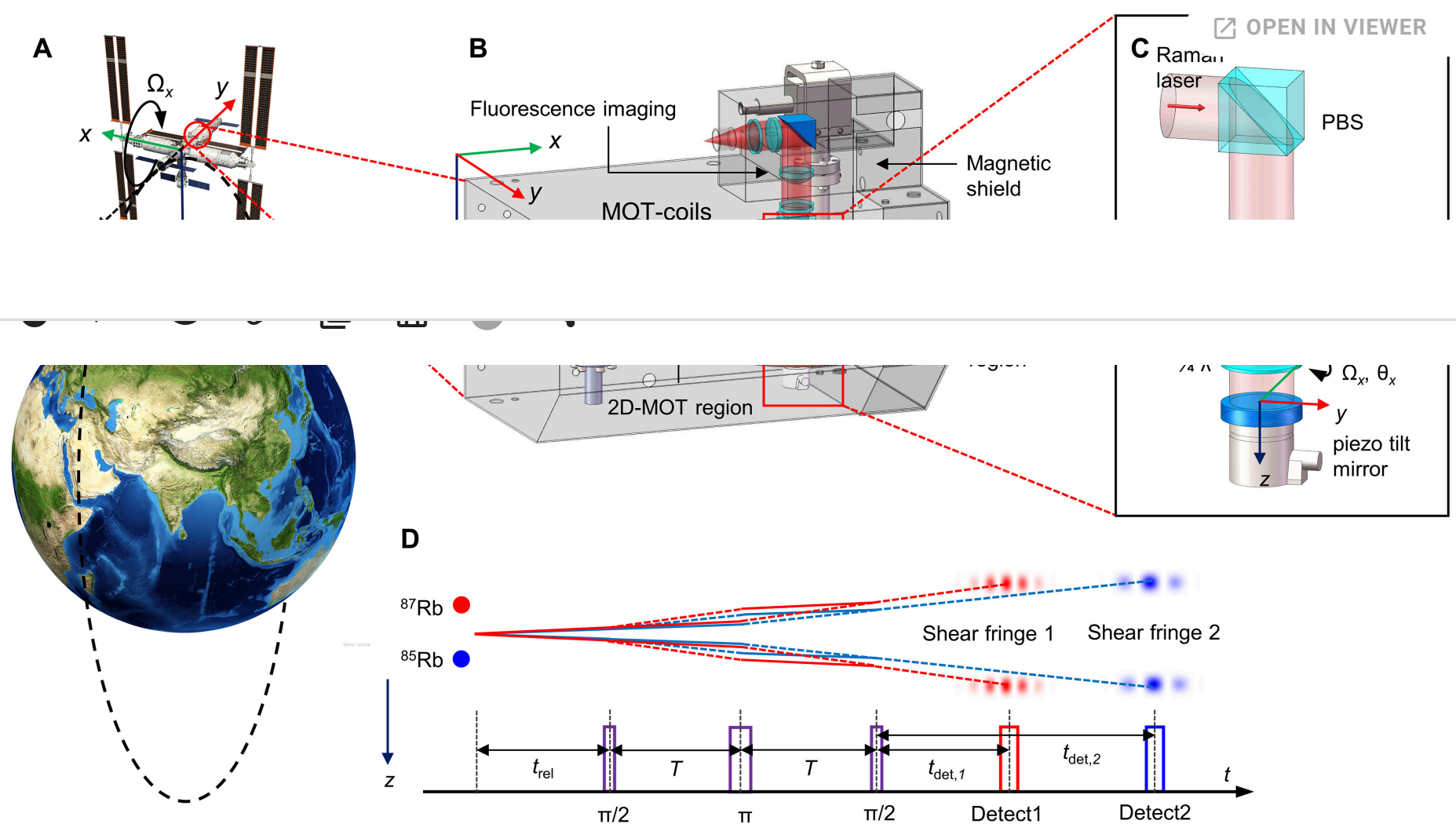


Fig. 1. Overview of the CSS Atom Interferometer (CSSAI).

(A) The CSS, its coordinate system, and its orbit around Earth. The CSSAI is installed inside the CSS. (B) The physical system of the CSSAI and its core functional modules. (C) Principle of point-source interferometry (PSI) using a piezo tilt mirror. The shearing fringe is induced by rotation rate of the CSS Ω_x and also the angle of the piezo tilt mirror θ_x . (D) Dual-species double single diffraction (DSD) interference sequence for ^{85}Rb and ^{87}Rb . The fluorescence of the isotopes is detected in sequence to obtain separate spatial interference fringes.

RESULTS

Experiment principle and procedure

The CSS Interferometer (CSSAI) is a compact, dual-species instrument installed within the CSS's High Microgravity Level Research Rack (HMLR) (44). Its core system (Fig. 1B) simultaneously prepares and coherently manipulates clouds of ^{85}Rb and ^{87}Rb atoms. The interferometer operates on the principle of double single diffraction (DSD) (Fig. 1D) (25), driven by counterpropagating Raman lasers that are reflected by a piezo tilt mirror. This configuration creates two symmetric interference loops per isotope. A critical feature of the setup is the point-source interference (PSI) scheme (Fig. 1C) (49–51), which transforms the interferometric phase into a spatial fringe pattern, al-

Raman laser are 440 and 425 MHz relative to the $^{85}\text{Rb } F = 3 \rightarrow F' = 4 \text{ D}_2$ transition and the $^{87}\text{Rb } F = 2 \rightarrow F' = 3 \text{ D}_2$ transition, respectively. The two-photon detuning is labeled as δ_{tp} . The width of the π Raman pulse is $\tau = 30 \mu\text{s}$. The fluorescence is captured through an imaging system with a magnification of 0.45 and recorded with a scientific complementary metal-oxide semiconductor camera. The fluorescence images of ^{85}Rb and ^{87}Rb would overlap if detected simultaneously. Therefore, we excite and detect their fluorescence sequentially at $t_{\text{det},1} = 40 \text{ ms}$ and $t_{\text{det},2} = 85 \text{ ms}$ relative to the last Raman pulse to obtain separate spatial interference images (Fig. 1D). By fitting the phases of the spatial fringes, the acceleration values along the z -direction are obtained (45).

The Eötvös parameter $\eta_{\text{Rb85,Rb87}}$ is defined as the ratio of the differential gravitational acceleration to the average gravitational acceleration between the ^{85}Rb and ^{87}Rb atoms induced by the Earth. In space, what we measure is the residual acceleration, which is the sum of the gravitational acceleration and the centrifugal acceleration, and the latter is the same for the dual-species atoms. Thus, $\eta_{\text{Rb85,Rb87}}$ is expressed as

$$\eta_{\text{Rb85,Rb87}} = \frac{a_{z,\text{Rb85}} - a_{z,\text{Rb87}}}{g_z} = \frac{\Delta\phi_\eta}{\bar{k}_{\text{eff}}g_zT^2} \quad (1)$$

where $a_{z,\text{Rb85}}$ and $a_{z,\text{Rb87}}$ are the residual accelerations measured by the ^{85}Rb and ^{87}Rb CAIs, g_z is the average gravitational acceleration of the dual-species atoms along the z direction, which is calculated from the altitude of the CSS and equals 8.72 m/s^2 . $\eta_{\text{Rb85,Rb87}}$ is derived from the WEP-violation-induced differential phase $\Delta\phi_\eta$ of the interference fringes, where $\bar{k}_{\text{eff}}$ is the averaged Raman laser's wave vector of the two Rb isotopes.

Platform motion effect suppression

The CSS maintains a nadir-pointing attitude with a constant rotation around the x axis (Fig. 1A). The x axis rotation Ω_x has a mean value of -1.138 mrad/s with a SD of $12.0 \mu\text{rad/s}$ over the long-term measurements, and the rotation rates in the other two axes are nearly zero. By precisely controlling the angles of the piezo tilt mirror during the Raman pulses according to an optimized relation (31, 45), the decoherent effect induced by the atom cloud's position distribution is eliminated, and proper fringe spatial periods are introduced for interference phase extraction. The optimized angle relation is related to the detection time t_{det} , and is different for the two Rb isotopes. We choose an effective detection time between $t_{\text{det},1}$ and $t_{\text{det},2}$ to calculate and set the optimized angles. Clear dual-species interference fringes are shown in Fig. 2 (A to D). The spatial frequencies f_{spa} of the observed fringe patterns are 1.72 and 1.36 rad/mm for ^{87}Rb and ^{85}Rb , respectively. This frequency difference arises from the different fluorescence detection times. Without angle compensation, the rotation of the CSS leads to a fringe frequency of about 0.5 rad/mm, which results in a phase variation less than 2π in the 1.1 cm image width. This makes it difficult to extract the interference fringes. The fringe contrast is also lower and the phase error is higher compared with the rotation-compensated case.

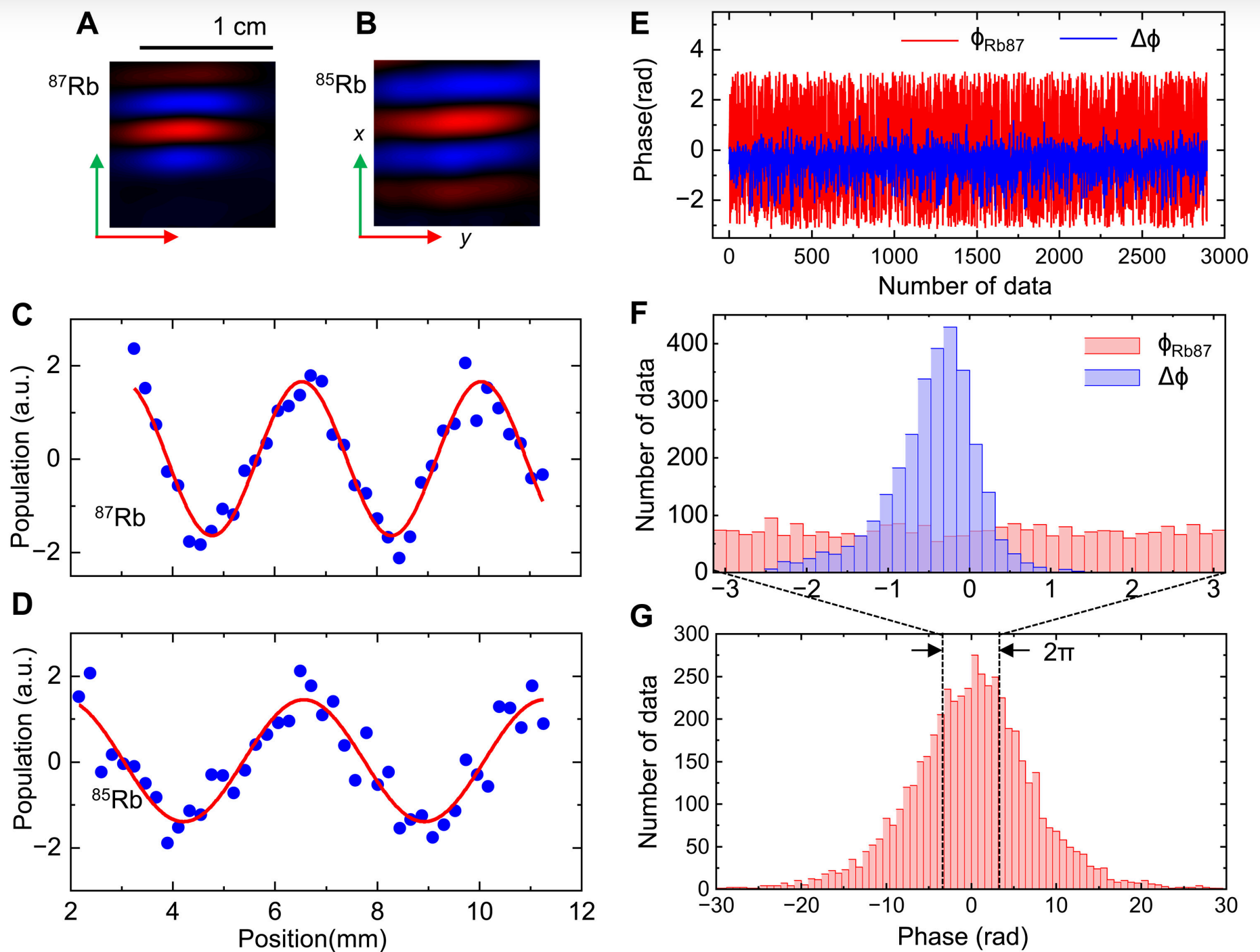


Fig. 2. Dual-species interference fringes and differential phase extraction.

(**A** and **B**) Principal component analysis (PCA) of the fluorescence images for (**A**), ^{87}Rb and (**B**), ^{85}Rb , showing 2D shearing fringes. Both are extracted from the 2D fluorescence images acquired over 366 experiments. (**C** and **D**) Typical 1D interference fringes for (**C**), ^{87}Rb and (**D**), ^{85}Rb , obtained by averaging the 2D images in the x direction and using a normalization method (45). (**E**) Time series of the measured interference phase ϕ_{Rb87} and the differential phase $\Delta\phi$. (**F**) Histograms of ϕ_{Rb87} and $\Delta\phi$. (**G**) Histogram of the acceleration induced interference phase calculated according to the acceleration recorded by a classical accelerometer.

The CSS's platform experiences residual acceleration due to the nonconservative force acting on the station, crew activities, and vibrations from scientific racks. The phases for a single-species atom interference fringes are measured, as shown in Fig. 2E. Their range is restricted to 2π due to the value range of the inverse sine function. The phase exhibits a uniform distribution, as shown in Fig. 2F, suggesting that its true distribution is much broader than 2π . For comparison, the acceleration of the CSSAI was measured by a classical accelerometer, and the corresponding phase was calculated from the measured acceleration using the sensitivity function (Supplementary Materials). The phase fluctuations for an individual interferometer are shown in Fig. 2G, and have a SD of 7.8 rad. The differential phase $\Delta\phi = \phi_{\text{Rb85}} - \phi_{\text{Rb87}}$ is measured and is shown in Fig. 2 (E and F). It has a SD of 0.55 rad, over an order of magnitude smaller than the single-species phase noise. This is because the dual-species interferometer, operating with an identical time sequence, inherently provides a high vibration rejection ratio on the order of $(k_{\text{eff,Rb87}} - k_{\text{eff,Rb85}})/\bar{k}_{\text{eff}} \sim 10^{-6}$. The residual noise of the differential phase is mainly attributed to the fringe's signal-to-noise ratio rather than the acceleration.

Suppression of the spatial frequency induced phase

The differential phase has a notable offset from zero, as shown in Fig. 2 (E and F). One reason for this offset is the spatial frequency mismatch of the interference fringes. The interference phases of the dual fringes are obtained through sine fitting. An offset of the initial fitting position x_0 will cause phase offset of $\phi = \pi x_0 / \lambda$. When the spatial fre-

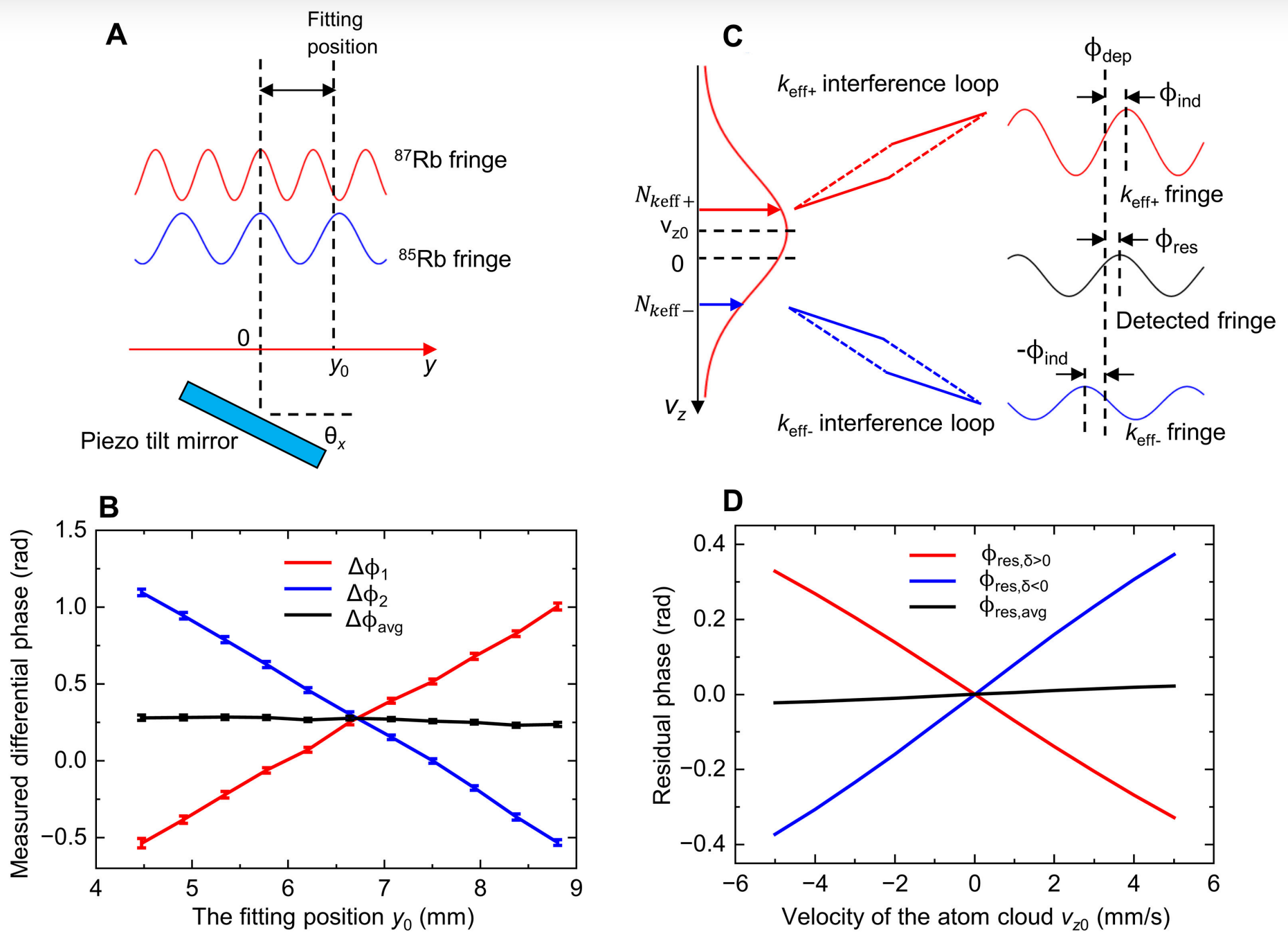


Fig. 3. Switching methods for fluorescence detection and two-photon detuning.

(A) Diagram of the induced differential phase when f_{spa} are different for the dual interferometers. (B) Calculated differential phase $\Delta\phi_i$ and their averaged $\Delta\phi_{\text{avg}}$ with varying sine-fitting offset y_0 for the interference fringe. (C) Diagram of the shearing fringe of $k_{\text{eff}\pm}$ and their summation when v_{z0} is not zero. (D) Calculation of the residual phase ϕ_{res} with $\delta_{\text{tp}} > 0$ and $\delta_{\text{tp}} < 0$ and their average with varying v_{z0} . ϕ_{ind} is set to 0.92 rad.

To solve this problem, we designed a fluorescence detection switching method. Two groups of dual-species interference experiments are conducted with alternated fluorescence detection sequence, then we measure the differential phase $\Delta\phi_i$ for each group and calculate their average $\Delta\phi_{\text{avg}}$, where $i = 1, 2$ represents different sequences. This method can effectively eliminate the phase shift induced by the spatial frequency mismatch. To verify its effectiveness, we fit the two groups of interference fringe with changing fitting position y_0 . The calculated $\Delta\phi_i$ and $\Delta\phi_{\text{avg}}$ are shown in Fig. 3B. The differential phase $\Delta\phi_i$ changes by approximately 1.6 rad when y_0 varies by 4.3 mm. However, the averaged differential phase $\Delta\phi_{\text{avg}}$ remains constant with a maximum phase change of 0.044 rad. A differential phase suppression ratio of about 36 is achieved.

We also performed a theoretical analysis to determine the magnitude and suppression ratio of the uncertainty associated with this phase offset. The formulas for $\Delta\phi_i$ and $\Delta\phi_{\text{avg}}$ are derived, and the uncertainties of these phases are calculated on the basis of typical values of the experimental parameters. Detailed information is illustrated in the “Analysis of the phase for the shearing fringe” section. The majority terms of the phase uncertainty of $\Delta\phi_i$ are induced by the uncertainty of y_0 and also the position uncertainty of the cold atom cloud. The calculated differential phase uncertainty for $\Delta\phi_i$ and $\Delta\phi_{\text{avg}}$ are 0.47 rad and 3.7×10^{-4} rad, respectively, corresponding to a suppression factor of more than 1000 by using the fluorescence detection switching method.

Suppression of the k_{eff} -independent phase

fringe is the sum of the fringes of the two loops and has a phase of ϕ_{det} . If the amplitudes of the two interference loops are the same, ϕ_{ind} is eliminated and ϕ_{det} is just ϕ_{dep} . However, if the amplitudes of the two interference loops are different, ϕ_{ind} will induce a residual phase shift ϕ_{res} , as shown in Fig. 3C.

The residual velocity of the atom cloud in the z -direction v_{z0} is the main factor that breaks the symmetry of the interference loops. A nonzero velocity component along the interferometer axis ($v_{z0} \neq 0$) results in different atom populations ($N_{k_{\text{eff}}\pm}$) being selected by the $k_{\text{eff}}\pm$ Raman transitions, as illustrated in Fig. 3C. A residual velocity of, e.g., 4 mm/s, would cause a 16% change in the population of the ^{85}Rb cold atom cloud. The value of v_{z0} cannot be measured directly for our experiment configurations, but its range is estimated to be within ± 4 mm/s, as estimated according to the measured velocity of the atom cloud in the x, y directions through the time-of-flight (TOF) method. Detailed illustration is in the Supplementary Materials.

To eliminate ϕ_{res} , we designed a two-photon detuning switching method. Two groups of dual-species interference experiments are conducted with alternated sign of the two-photon detuning δ_{tp} . $N_{k_{\text{eff}}\pm}$ selected by the $k_{\text{eff}}\pm$ transitions are then reversed for these two configurations, and the signs of ϕ_{res} are reversed too. An average of the detected phases ϕ_{det} for these two configurations will eliminate the residual phase ϕ_{res} . Numerical simulation of the DSD interference process for the PSI fringe (“Simulation for the PSI fringe” section) is developed to calculate the phase error elimination ratio of this method. We set the k_{eff} -independent phase ϕ_{ind} to be 0.92 rad, and vary v_z from -5 to 5 mm/s. Then, we calculate ϕ_{res} for $\delta_{\text{tp}} > 0$ and $\delta_{\text{tp}} < 0$ and their average according to the actual experiment parameters of ^{87}Rb CAI. The results are shown in Fig. 3D. The residual phases for individual configurations exhibit opposite slopes, with a maximum amplitude of 0.65 rad. However, the averaged residual phase has a maximum amplitude of 0.04 rad, which shows a residual phase suppression ratio about 15.

Long-term differential phase measurement

Using the two switching methods, we conducted a long-term WEP test with four experimental configurations. Over a period of 280 days, we acquired more than 9700 pairs of interference fringes. The differential phases of each configuration are averaged to 44 data points to reduce the noise. The averaged differential phases are denoted as $\overline{\Delta\phi}_{i,j}$, where $i = 1, 2$ represents different sequences, and $j = +, -$ represents different sign of δ_{tp} . The averaged phase $\overline{\Delta\phi}_{\text{avg}} = \sum_{i,j} \overline{\Delta\phi}_{i,j} / 4$ are also calculated. The calculated phases are shown in Fig. 4A. The differential phase bias is reduced from a maximum value of -0.556 rad for individual configurations to an average value of -0.117 rad, demonstrating the effectiveness of the switching methods and its critical role in obtaining a reliable WEP test result.

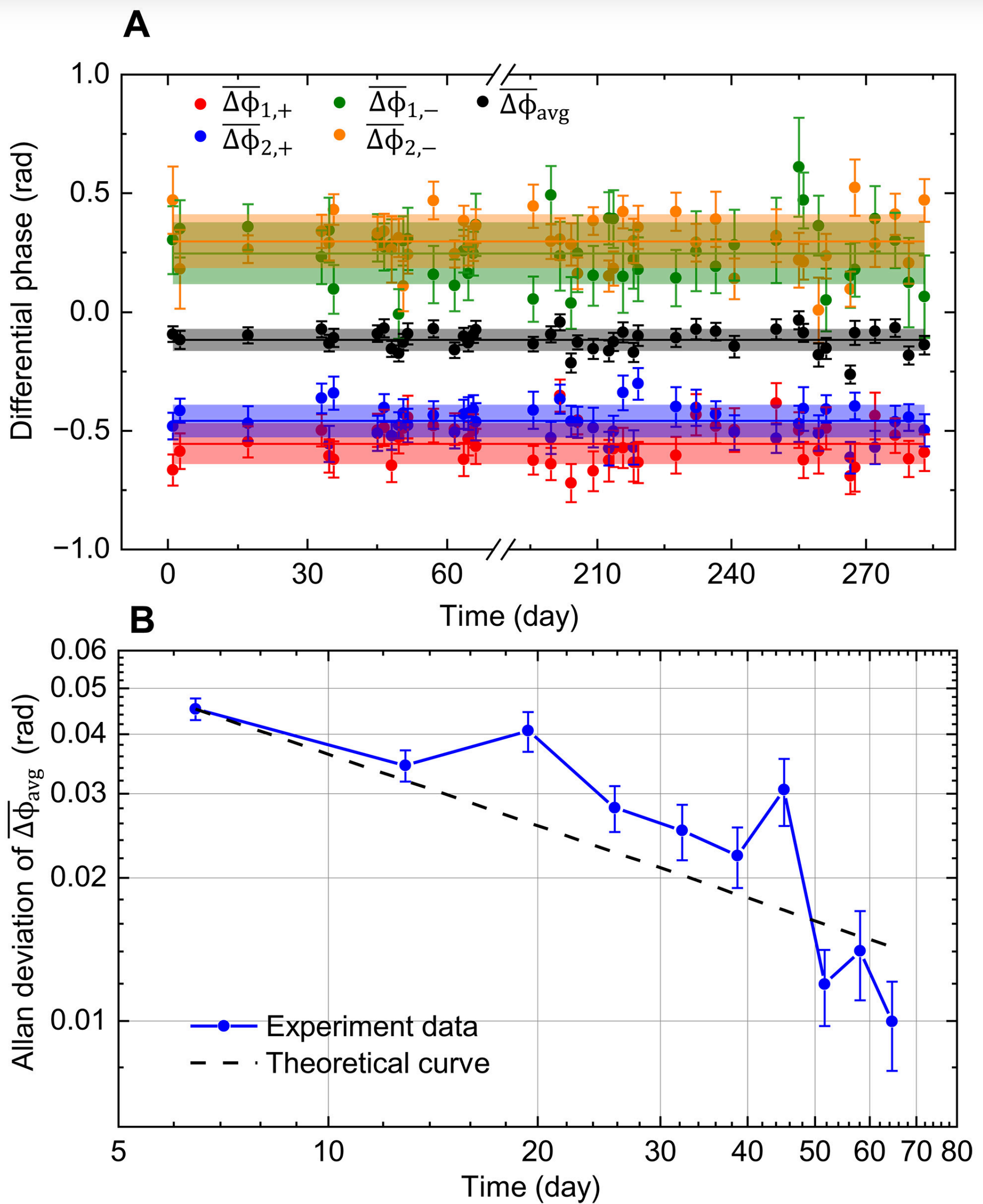


Fig. 4. Long-term differential phase measurement and stability.

(A) Averaged differential phase $\Delta\phi_{i,j}$ for the four measurement configurations and their average $\Delta\phi_{avg}$ over long-term measurement. The starting date is 27 August 2024. The dots represent the values of experiment data. The solid lines represent the data's average values. The dashed areas represent the data's SD. (B) Allan deviation of $\Delta\phi_{avg}$ over the experiment time t , demonstrating the stability of the differential phase measurement. The dots represent the calculated Allan deviation. The dashed line represents the theoretical curve. It is calculated by dividing the Allan deviation of the first data point by $\sqrt{t/t_{sin}}$, where t_{sin} is the effective time interval of a single differential phase.

The averaged differential phases $\Delta\phi_{i,j}$ exhibit a SD of 0.046 rad. The long-term stability of the differential phase is

Error estimation

A detailed error estimation is essential for obtaining an accurate WEP test result. The major systematic error terms are summarized in [Table 1](#). In [Fig. 2F](#), we observe skewness in the differential phase distribution. This is because we fit a noisy fringe for both its spatial frequency and phase, which leads to an asymmetric distribution of the extracted phase. Through simulation, we find that the average of the fitted phase has an offset from its actual value. A correction is therefore applied to estimate the differential phase more accurately. Besides the phase error induced by data processing, the dominant uncertainty arises from a newly identified effect: imaging angle offset, which results from the coupling between the centroid offset of the DSD interference fringes and the angular misalignment of the imaging system. The centrifugal force and gravity gradient effects are both linked to the centroid offset of the DSD interference loops, which is primarily induced by the residual velocity v_{z0} . The single-photon ac Stark shift introduces a k_{eff} -independent phase and is effectively suppressed by the two-photon detuning switching method. Phase shifts induced by residual acceleration, the multiple-sideband effect of the Raman lasers, and uncertainties in k_{eff} and T are all proportional to the local gravitational acceleration, which are suppressed by at least three orders of magnitude compared to ground-based measurements. A detailed discussion of each systematic effect is provided in the “Error estimation” section. The measured average differential phase $\overline{\Delta\phi}_{\text{avg}}$ after error correction is (-0.10 ± 0.17) rad. The Eötvös parameter $\eta_{\text{Rb85,Rb87}}$ is then calculated to be $(-2.7 \pm 4.7) \times 10^{-7}$ according to [Eq. 1](#).

Terms	Differential phase (rad)
Measurement result	-0.12 ± 0.01

Expand for more ▼

Table 1. Error budget for the differential phase.

DISCUSSION

This paper presents the WEP test experiment conducted with the CSSAI. This work successfully transitions such experiments from short-duration microgravity platforms to a permanent orbital laboratory. The instrument operated stably for over 280 days. We addressed the challenges posed by the rotation rate and residual acceleration of the CSS platform. Methods were designed to suppress the differential phase errors. Error estimations are carried out to give a test result with a precision of 4.7×10^{-7} , improving upon the best previous microgravity test result based on atom interferometry by three orders of magnitude.

The methods developed in this paper are also beneficial for future space-based CAI WEP tests. By rotating the spacecraft with a certain spin velocity, one can modulate the WEP signal and suppress the noise at the orbital frequency (4). For this case, the proposed PSI interference scheme with optimized angle relation could be used to extract the interference phase and reduce the atom parameter induced phase error. The proposed two-photon detuning switching method eliminates the k_{eff} -independent phase and can improve the measurement accuracy for the Raman or single-photon diffraction interference scheme (52). This study also provides a comprehensive error analysis, identifying several newly discovered sources of phase error and highlighting differences between ground-based and space-based experiments. This provides a reference for future space-based cold atom interferometer design and error analysis.

for quantum precision measurement in space, this work represents a substantial step toward the practical application of space-based CAI, opening a new avenue for fundamental physics research in space.

MATERIALS AND METHODS

Experimental design

The CSSAI operates in an intermittent mode on the free-floating platform for microgravity experiment within the HMLR (45). For a single round of WEP test experiments, the total operation time is limited to 70 min to prevent overheating of the payload. The first 13 min are dedicated to the startup process, which includes stabilization of temperature control, achievement of a stable vacuum level, and automatic locking of the laser frequencies. The subsequent 18 min are allocated to scientific experiments, during which the atom interference sequence is executed periodically. WEP test experiments with four experimental configurations are performed alternately, and approximately 30 pairs of interference fringes are obtained per configuration per round. The final 39 min are reserved for data transfer.

Analysis of the phase for the shearing fringe

Phase of the PSI interference fringe

The formula for shearing fringe when considering the acceleration, rotation and Raman mirror's angle is (45)

$$\phi = k_{\text{eff}} a_z T^2 + k_{\text{eff}} [2\Omega_x v_y T^2 + \theta_{x,1} r_y + \theta_{x,3} (r_y + 2v_y T)] \quad (2)$$

where k_{eff} is the effective wave vector, T is the time interval between the Raman pulses, a_z is the residual acceleration along the z axis, Ω_x is the rotation rate of the CSS around the x axis, and $\theta_{x,1}$ and $\theta_{x,3}$ are the angles of the Raman laser beam along the x axis at the time of the first and third Raman laser pulses, relative to its angle at the time of the second Raman laser pulse. r_y and v_y are the position and velocity of the atom at the time of the 1st Raman laser pulse.

When $\theta_{x,1}$ and $\theta_{x,3}$ have the following relationship (45),

$$\theta_{x,1} = \theta_{x,3} = \frac{-t_{\text{det}}\theta_{x,3} + 2\Omega_x T^2}{2T + t_{\text{det}}} \quad (3)$$

the decoherent effect induced by the atom cloud's distribution is eliminated, where t_{det} is the time delay between the last Raman laser pulse and the detection laser pulse. The shearing fringe has the following form (45)

$$\phi = f_{y0} R_y \quad (4A)$$

$$f_{y0} = \frac{2k_{\text{eff}}}{2T + t_{\text{det}}} (\theta_{x,3} T + \Omega_x T^2) \quad (4B)$$

where $R_y = r_y - v_y(2T + t_{\text{det}})$ is the position of the atom at the detection time. f_{y0} is the optimized spatial frequency. When the relation of Eq. 3 is not exactly fulfilled and has an angle offset $\Delta\theta_x = \theta_{x,1} - \theta_{x,3}$, then both the phase and the spatial frequency have to be modified.

where $\alpha = \frac{t_{\text{tot}} - t_{\text{rel}}}{t_{\text{tot}}} \cdot \frac{\sigma_{py}^2}{\sigma_{vy}^2 t_{\text{tot}}^2 + \sigma_{py}^2}$ and $\beta = \sigma_{vy}^2 t_{\text{tot}}^2 / \sigma_{py}^2$, t_{rel} is the time from atom release to the first Raman pulse, $t_{\text{tot}} = t_{\text{rel}} + 2T + t_{\text{det}}$ is the total time. σ_{py} , σ_{vy} are the distribution widths of the position and velocity of the atom cloud. ρ_{y0} and v_{y0} are the initial position and initial velocity of the atom cloud.

Differential phase of the PSI interference fringe

For the dual-species atom interferometer WEP test experiment, the fluorescence detection times for the two isotopes are different, denoted as $t_{\text{det},1}$ and $t_{\text{det},2}$. This will lead to different ideal angles $\theta_{x,1}$ according to Eq. 3. Therefore, we use a time t_{det} midway between $t_{\text{det},1}$ and $t_{\text{det},2}$ to set the angle of $\theta_{x,1}$ according to Eq. 3. Then, the modified phase of Eq. 5 is denoted as $\phi(i, j)$, where $i = 1, 2$ for ^{85}Rb and ^{87}Rb isotopes and $j = 1, 2$ for detection times $t_{\text{det},1}$ and $t_{\text{det},2}$. The differential phase $\Delta\phi_1$ for the case where ^{85}Rb is detected before ^{87}Rb , and $\Delta\phi_2$ for the case where ^{87}Rb is detected before ^{85}Rb are

$$\Delta\phi_1 = \phi(1, 1) - \phi(2, 2) \quad (6A)$$

$$\Delta\phi_2 = \phi(1, 2) - \phi(2, 1) \quad (6B)$$

After derivation, $\Delta\phi_1$ has the form

$$\begin{aligned} \Delta\phi_1 = & k_{\text{eff}} [\alpha_{1,1} (\beta_{1,1} \rho_{1,y0} - v_{1,y0} t_{\text{tot},1}) \Delta\theta_{x,1} - \alpha_{2,2} (\beta_{2,2} \rho_{2,y0} - v_{2,y0} t_{\text{tot},2}) \Delta\theta_{x,2}] \\ & + k_{\text{eff}} \Delta f_y y_0 + k_{\text{eff}} [(t_{\text{rel}}/t_{\text{tot},1} + \alpha_{1,1}) \Delta\theta_{x,1} - (t_{\text{rel}}/t_{\text{tot},2} + \alpha_{2,2}) \Delta\theta_{x,2}] y_0 \end{aligned} \quad (7)$$

where, $\alpha_{i,j}$, $\beta_{i,j}$, $\rho_{i,y0}$, $v_{i,y0}$, $t_{\text{tot},j}$, $\Delta\theta_{x,j}$ are defined parameters with $i = 1, 2$ for ^{85}Rb and ^{87}Rb isotopes and $j = 1, 2$ for detection times $t_{\text{det},1}$ and $t_{\text{det},2}$. Δf is the differential spatial frequency with $\Delta f_y = f_{y,1} - f_{y,2}$. The zero point of R_y is at the position of the rotation axis of the tilt mirror, and y_0 is the fitting position of the shearing fringe from this zero point, as shown in Fig. 3A. $\Delta\phi_2$ has similar form.

The average differential phase $\Delta\phi_{\text{avg}}$ for $\Delta\phi_1$ and $\Delta\phi_2$ are

$$\Delta\phi_{\text{avg}} = (\Delta\phi_1 + \Delta\phi_2)/2 \quad (8)$$

After derivation, $\Delta\phi_{\text{avg}}$ has the form

$$\begin{aligned} \Delta\phi_{\text{avg}} = & \frac{1}{2} k_{\text{eff}} \{ [\alpha_{1,1} (\beta_{1,1} \rho_{1,y0} - v_{1,y0} t_{\text{tot},1}) - \alpha_{2,1} (\beta_{2,1} \rho_{2,y0} - v_{2,y0} t_{\text{tot},1})] \Delta\theta_{x,1} \\ & + [\alpha_{1,2} (\beta_{1,2} \rho_{1,y0} - v_{1,y0} t_{\text{tot},2}) - \alpha_{2,2} (\beta_{2,2} \rho_{2,y0} - v_{2,y0} t_{\text{tot},2})] \Delta\theta_{x,2} \} \\ & + [(\alpha_{1,1} - \alpha_{2,1}) \Delta\theta_{x,1} + (\alpha_{1,2} - \alpha_{2,2}) \Delta\theta_{x,2}] y_0 \end{aligned} \quad (9)$$

Two features can be seen for $\Delta\phi_{\text{avg}}$. First, the phase term $k_{\text{eff}} \Delta f_y y_0$ is eliminated due to the fluorescence detection switching. Second, if the two atom clouds have identical distributions, that is $\alpha_{1,j} = \alpha_{2,j}$, $\beta_{1,j} = \beta_{2,j}$, $\rho_{1,y0} = \rho_{2,y0}$, $v_{1,y0} = v_{2,y0}$, then the phase error of $\Delta\phi_{\text{avg}}$ is completely suppressed according to Eq. 9. If the distributions of the atom clouds are not the same, residual phase error exists.

Error estimation for the differential phase

differential phases are calculated and are shown in Table 2. The calculated value for $\Delta\phi_1$ is (0.00 ± 0.47) rad. The majority uncertainties come from the ρ_{y0} and y_0 , where the spatial frequency difference couples to the position uncertainty and contributes phase uncertainty. The calculated value for $\Delta\phi_{\text{avg}}$ is $(-1.5 \pm 3.7) \times 10^{-4}$ rad. By applying the fluorescence detection switching method, the uncertainty of the differential phase is suppressed by over 1000 times.

Error source	Parameters	$\Delta\phi_1$ (rad)	$\Delta\phi_{\text{avg}}$ (rad)
Phase offset		1.2×10^{-3}	-1.5×10^{-4}

[Expand for more](#)

Table 2. Error estimation of the differential phase $\Delta\phi_1$ and $\Delta\phi_{\text{avg}}$ based on actual experimental parameters.

Simulation for the PSI fringe

To calculate phase shifts induced by effects that are difficult to analyze theoretically, we developed a simulation program for the interference process. This program starts from the initial state of the atom with velocity distribution, solves the state evolution using the Schrödinger equation with spatial coupled phase such as the rotation and tilted angles induced phase, and computes the shearing interference fringes in the position space. On the basis of the obtained fringes, the phase shifts induced by various parameters can be calculated. Since the interference processes for the two atomic species are similar, we introduce the procedure of the simulation for one species.

State evolution equation and Hamiltonian

The configuration of the Raman lasers for atom interferometry is shown in Fig. 5A. I_1 and I_2 are the incident Raman lasers. They are linearly polarized with the same polarization direction. I_1 and I_2 are reflected with the piezo tilt mirror. A quarter-wave plate before the mirror adjusts the Raman laser polarization so that the reflected and incident beams form pairs with perpendicular linear polarizations. The reflected lasers are labeled as $I_{1,r}$, $I_{2,r}$. I_1 and $I_{2,r}$ form the $k_{\text{eff}+}$ Raman laser pair, and I_2 and $I_{1,r}$ form the $k_{\text{eff}-}$ Raman laser pair.

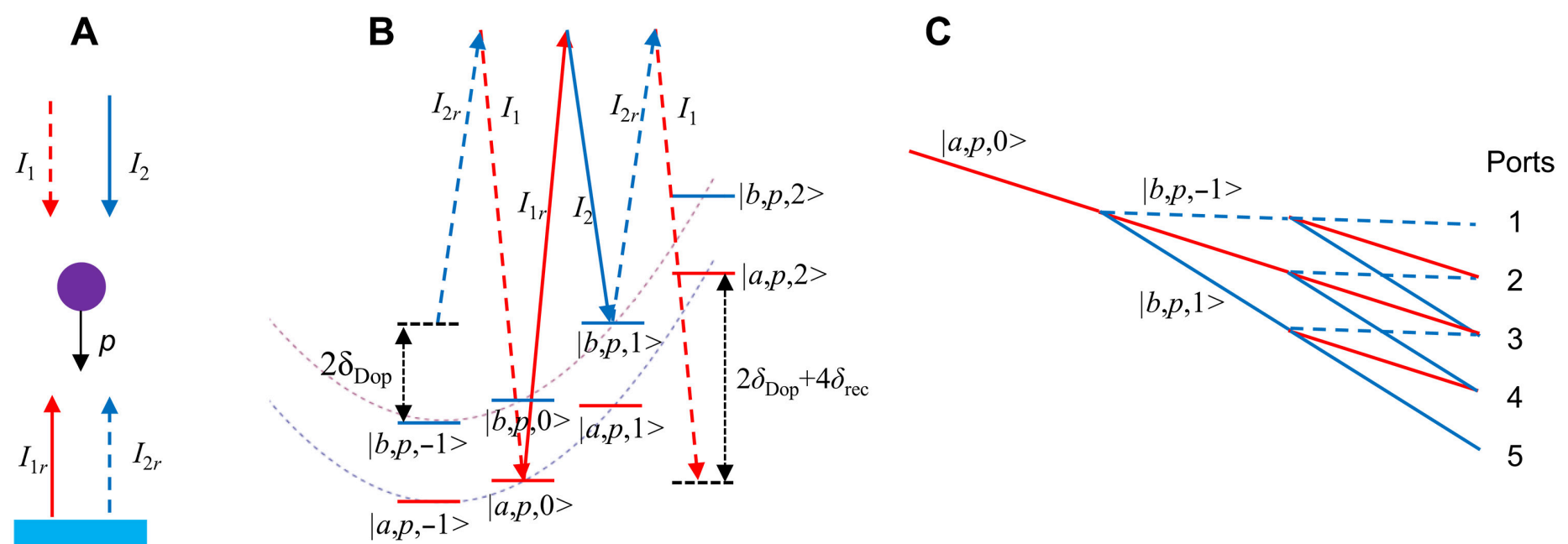


Fig. 5. Raman lasers and interferometer dynamics for the simulation model.

(A) The counterpropagating Raman laser pairs. (B) Momentum states coupled by the DSD Raman transitions. (C) Interference loops involving the $|a,p,0\rangle$ and $|b,p,-1\rangle$ states.

states labeled $|a, n\hbar k_{\text{eff}} + p\rangle$ and $|b, n\hbar k_{\text{eff}} + p\rangle$, where $n = 0, \pm 1, \dots$ and $p = mv$, as shown in Fig. 5B. For convenience, we denote them as $|a, p, n\rangle$ and $|b, p, n\rangle$. The state amplitudes of the above states are denoted as $c(a, p, n)$ and $c(b, p, n)$.

Following the derivation in (57), the state amplitudes can be calculated by the coupled equations

$$i\hbar\dot{c}(b, p, n) = \frac{\hbar\Omega_{\text{eff}+}}{2}\exp^{i[(2n-1)\delta_{\text{rec}}-\delta_{\text{Dop}}-\delta_{\text{tp}}]t-i\phi_{\text{eff}+}}c(a, p, n-1) + \frac{\hbar\Omega_{\text{eff}-}}{2}\exp^{i[(-2n-1)\delta_{\text{rec}}+\delta_{\text{Dop}}-\delta_{\text{tp}}]t-i\phi_{\text{eff}-}}c(a, p, n+1) \quad (10A)$$

$$i\hbar\dot{c}(a, p, n) = \frac{\hbar\Omega_{\text{eff}-}}{2}\exp^{-i[(-2n+1)\delta_{\text{rec}}+\delta_{\text{Dop}}-\delta_{\text{tp}}]t+i\phi_{\text{eff}-}}c(b, p, n-1) + \frac{\hbar\Omega_{\text{eff}+}}{2}\exp^{-i[(2n+1)\delta_{\text{rec}}-\delta_{\text{Dop}}-\delta_{\text{tp}}]t+i\phi_{\text{eff}+}}c(b, p, n+1) \quad (10B)$$

where, $\Omega_{\text{eff}\pm}$ are the effective Rabi frequency of the $k_{\text{eff}\pm}$ Raman transition, $\phi_{\text{eff}\pm}$ are the Raman laser's phases, δ_{rec} is the recoil frequency, $\delta_{\text{Dop}} = k_{\text{eff}}v$ is the Doppler frequency shift. Eq. 10 could be expressed as

$$i\hbar\dot{\varphi}(t) = H\varphi(t) \quad (11)$$

where $\varphi(t) = \sum_n c(a, p, n) |a, p, n\rangle + c(b, p, n) |b, p, n\rangle$.

Calculation of the PSI interference fringe

The Mach-Zehnder interferometer sequence consists of $\pi/2 - \pi - \pi/2$ Raman pulses. The end state through these pulses is calculated using

$$\varphi_{\text{end}}(t) = U_{\pi/2}U_{\pi}U_{\pi/2}\varphi_{\text{int}}(t) \quad (12)$$

where $U = e^{-iH/\hbar}$ is the evolution operator.

Equation 12 describes the internal state evolution. However, it does not consider the decoherence effect between different momentum states. Figure 5C shows the interference loops when considering the $|a, p, 0\rangle$ and $|b, p, \pm 1\rangle$ states. There are five exit ports for the interference loop. If Eq. 12 alone is used to calculate the final state, coherence is assumed between states at different exit ports, as shown in Fig. 5C. However, the finite temperature of the atom cloud causes decoherence between states from different ports. To address this issue, we use a state-grouping method to calculate the population for each exit port individually (57). We calculate the state populations for each port, and then calculate the total population over all ports.

To simulate the PSI spatial fringe, the velocity distribution of the atom cloud in the y and z directions is considered. The velocity distribution is discretized into individual velocity component. For each discrete velocity component in the initial state, Eq. 12 associated with the grouping method is used to calculate the population of the end state. The shearing phases described in Eq. 2 is added to the Raman laser's phase during the calculation. The 2D PSI interference fringes are a sum of the end state population over all velocity components. Typical fringes in the y-z plane are shown in Fig. 6 (A and B). A pair of spatial fringes in the y direction and separated along the z direction are clearly shown.

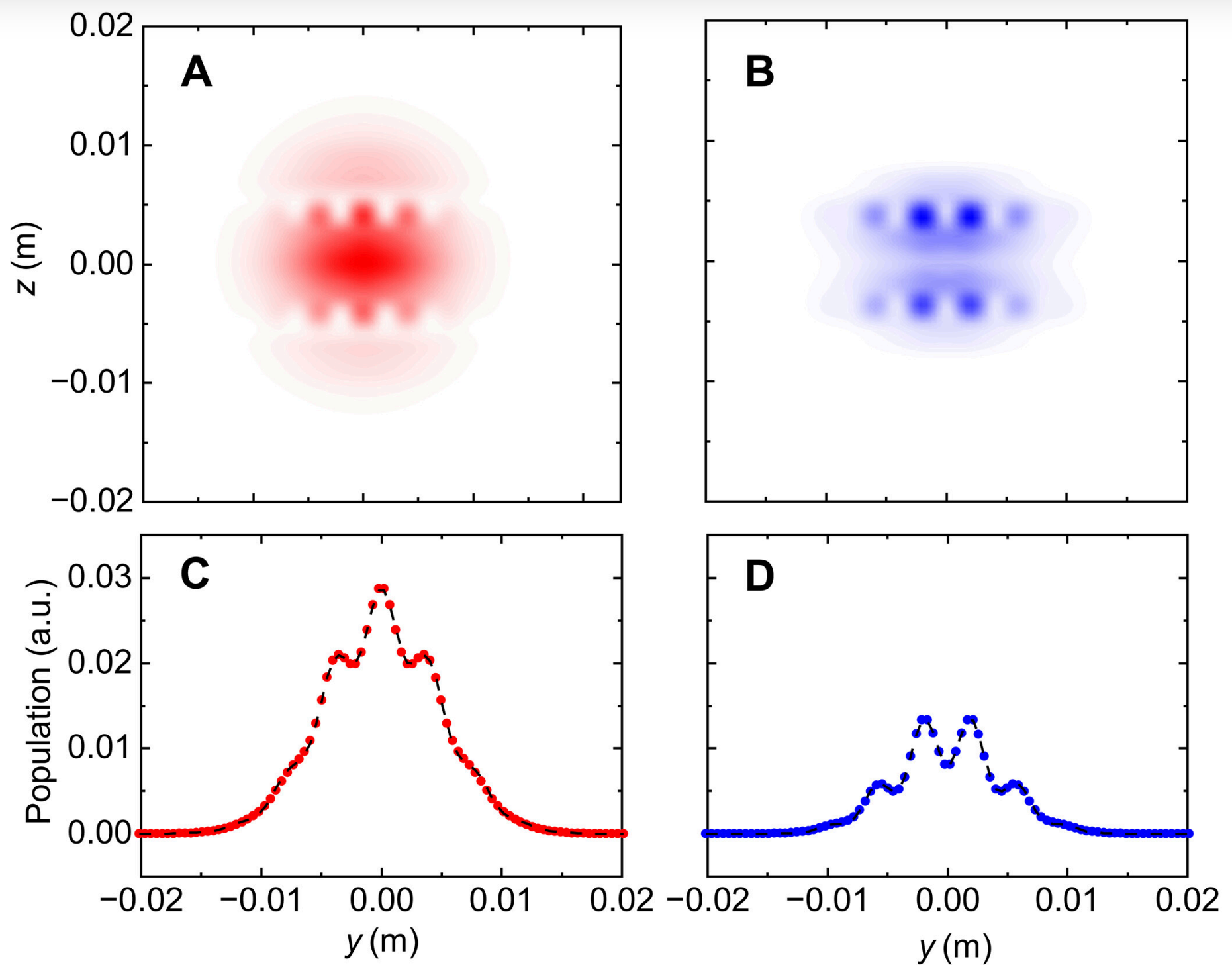


Fig. 6. Simulation result of the point-source interferometer (PSI) fringes.

(**A** and **B**) Simulated 2D interference fringes in the y - z plane for the end states $|a\rangle$ and $|b\rangle$. (**C** and **D**) The corresponding 1D spatial fringes obtained by integrating the 2D population distributions along the z direction. The simulation parameters are: atom cloud temperature $6.7\ \mu\text{K}$, $t_{\text{rel}} = 43\ \text{ms}$, $T = 50\ \text{ms}$, $t_{\text{det}} = 40\ \text{ms}$, $\tau = 30\ \mu\text{s}$, and $\delta_{\text{tp}} = 69\ \text{kHz}$. $\Omega_x = -0.0011\ \text{rad/s}$, $\theta_{x,1} = -93\ \mu\text{rad}$, and $\theta_{x,3} = 188\ \mu\text{rad}$. The initial velocity of atom cloud v_{z0} is set to $4\ \text{mm/s}$.

To extract the fringe phase, the 2D population is averaged along the z -direction to obtain the 1D shearing fringe, as shown in **Fig. 6 (C and D)**. By fitting these curves with a model of Gaussian envelope modulated by sinusoidal fringe, we can calculate the phase and also other parameters such as the fringe contrast and spatial frequency. Phase shifts induced by various parameters, such as AC Stark shift, atom cloud velocity, rotation, etc., can be calculated.

Error estimation

Phase distribution skewness

In the fringe fitting process, we simultaneously fit the spatial frequency and phase. We performed simulations (see the Supplementary Materials). For simulated fringes with zero phase and spatial frequency and signal-to-noise ratio similar to those of the actual fringes, notable skewness is observed in the fitted phase distribution. The average and median of the fitted phase from the simulated fringes are $(31 \pm 5)\ \text{mrad}$ and $(1 \pm 5)\ \text{mrad}$, respectively. The median phase provides a more accurate estimate of the true phase. We therefore adopt the median phase as the estimated phase and take the difference between the median and the average as the phase error. We then calculated the average and median differential phases and their differences for the actual data from the four experimental configurations used in the WEP test, yielding a differential phase correction of $(0.01 \pm 0.03)\ \text{rad}$, as listed in **Table 1**.

offset θ_z in the z direction, it will induce a phase offset of $\phi = f_{\text{spa}}\theta_z z_{\text{fri}}$. For the dual-interference fringe, a centroid offset difference Δz_{fri} will cause a differential phase offset, $\Delta\phi = \bar{f}_{\text{spa}}\theta_z \Delta z_{\text{fri}}$, where $\bar{f}_{\text{spa}}$ is the averaged spatial frequency. This phase shift could not be suppressed by the proposed switching methods. Considering the uncertainty of the sign of v_{z0} , and the installation uncertainty of the imaging system, the estimated differential phase is (0.00 ± 0.16) rad. Detailed illustration is in the Supplementary Materials.

Imaging angle offset in the x - y plane will also induce differential phase error. The initial velocities of the dual atom clouds in the x - y plane are not exactly the same due to the imperfections of the experimental parameters. It will cause an x -direction difference Δx of the dual-species interference fringes center. If the imaging system has a rotation angle θ_{rot} in the x - y plane, it will cause a differential phase offset $\Delta\phi = \bar{f}_{\text{spa}}\theta_{\text{rot}}\Delta x$. θ_{rot} is measured through the 2D interference fringe and is found to be $\theta_{\text{rot}} = (-0.034 \pm 0.011)$ rad. The value of Δx is measured through a method that rotates the image angle of the 2D fringe image in x - y plane, and is found to be -0.175 mm. Detailed illustration is in the Supplementary Materials. The induced differential phase is calculated to be (0.009 ± 0.003) rad. Taking into account the two imaging misalignment effects, the induced differential phase is (0.01 ± 0.16) rad.

Acceleration

The differential phase arising from residual acceleration is given by $\Delta\phi = \Delta k_{\text{eff}} a_z T^2$, where Δk_{eff} is the differential effective wave vector and a_z denotes the residual acceleration measured by a classical accelerometer. Δk_{eff} is six orders lower than k_{eff} , which means the WEP test precision induced by the residual acceleration is also suppressed by the same order. The acceleration is measured by a classical accelerometer. The measured residual acceleration a_z is $(2.7 \pm 0.6) \times 10^{-4} \text{ m/s}^2$ during the WEP test experiment, and the accelerometer has a zero bias of $\pm 10^{-3} \text{ m/s}^2$. The synthetic residual acceleration value is $(2.7 \pm 10.0) \times 10^{-4} \text{ m/s}^2$, and its induced differential phase is $(-2.1 \pm 8.0) \times 10^{-5}$ rad.

Rotation and piezo tilt mirror

Rotation and the tilt angle of the piezo tilt mirror can induce a residual differential phase, even after applying the detection switching method. The derived formula for their influence is illustrated in the “Experimental design” section. By substituting actual experimental parameters, the calculated contribution of $\overline{\Delta\phi}_{\text{avg}}$ for this effect is $(-1.5 \pm 3.7) \times 10^{-4}$ rad. The differential phase uncertainty is mainly caused by the uncertainties of the position of the atom clouds and the fit position y_0 relative to the position of the tilt mirror’s rotation axis. The rotation will also cause centrifugal force. The induced differential phase is $\Delta\phi = -k_{\text{eff}}\omega_x^2 \Delta z_{\text{int}} T^2$, where $\Delta z_{\text{int}} = z_{\text{int,Rb85}} - z_{\text{int,Rb87}}$ is the centroid position difference of the ^{85}Rb and ^{87}Rb interference loops. This centroid position offset is mainly caused by the v_{z0} induced interference loop asymmetry. If v_{z0} is 4 mm/s, the calculated z_{int} is 0.7 mm. Considering the uncertainty of v_{z0} of the two cold atom clouds, the induced differential phase is $(0.0 \pm 7.4) \times 10^{-5}$ rad.

Single-photon light shift

The CSSAI uses electro-optic phase modulation to generate laser sidebands that serve as the Raman lasers (58). The ratios of the sidebands are monitored by an F-P cavity. According to the measured sideband ratios, the total power of the Raman lasers, and the single-photon detuning of the Raman lasers, the single-photon light shifts δ_{spls} are calculated to be -27.7 and 38.4 kHz for ^{85}Rb and ^{87}Rb . These frequency shifts were verified by in-orbit Raman transition spectroscopy measurements.

δ_{spls} induced phase is k_{eff} -independent for each interference loop of DSD. It causes residual phase ϕ_{res} only when the interference loops are asymmetric. The residual phase is calculated to be proportional to δ_{spls} and v_{z0} . Because δ_{spls} of the two isotopes have opposite signs, the contribution of $\overline{\Delta\phi}_{\text{avg}}$ induced by this effect will be maximum when v_{z0} of the two atom clouds have the same sign. By applying the two-photon detuning switching method, the induced differential phase could be suppressed to some extent.

tial phases are 0.057 and -0.076 rad for the two cases. The induced averaged differential phase is -0.01 rad. Considering the uncertainty of the sign of v_{z0} , the induced phase for $\overline{\Delta\phi_{\text{avg}}}$ is $(0.0 \pm 1.0) \times 10^{-2}$ rad.

Two-photon light shift

The two-photon light shift δ_{tpls} arises from off-resonant Raman transitions. For the DSD interference process, the two-photon detuning is nonzero. When the $k_{\text{eff}+}$ Raman transition is in resonance to an atom ensemble with a certain velocity distribution, the $k_{\text{eff}-}$ Raman transition is off-resonance to this atom ensemble, and vice versa. δ_{tpls} is calculated through the effective Rabi frequency and the two-photon detuning (59). At the free fall condition on ground, δ_{tpls} induced phase is mainly induced by the gravity induced Doppler frequency shift, and is a kind of k_{eff} -dependent phase (59). However, under microgravity condition, the gravity-induced Doppler frequency changing is absent, δ_{tpls} induced phase is found to be k_{eff} -independent. Similar to the single-photon light shift case, the induced residual phase exists only when the DSD interference loop is asymmetric. We set v_{z0} for both atom clouds to have a magnitude of 4 mm/s but with opposite sign to maximize the induced differential phase. For the cases of positive and negative two-photon detuning, the calculated differential phases are -2.0 and -2.6 mrad. Considering the uncertainty of the sign of v_{z0} , the induced differential phase is $(0.0 \pm 2.3) \times 10^{-3}$ rad.

Magnetic field

For the DSD interference, the magnetic field induced phase shift is $\phi = 2\pi \alpha B_0 \gamma \hbar k_{\text{eff}} T^2 / m_{\text{Rb}}$, where α is the second order Zeeman frequency shift coefficient, $B_0 = 504$ mGal is bias magnetic field, γ is its gradient, and m_{Rb} is the mass of rubidium. γ is evaluated on ground and has an uncertainty of (0.0 ± 5.6) mG/m (45). This phase shift is k_{eff} dependent. The induced differential phase is $(0.0 \pm 3.8) \times 10^{-4}$ rad.

Other effects

The gravity gradient induced differential phase is $\Delta\phi = k_{\text{eff}} T_{zz} \Delta z_{\text{int}} T^2$, where T_{zz} is the zz component of the gravity gradient tensor, which has a value of $(-2.582 \pm 0.017) \times 10^{-6} \text{ s}^{-2}$ according to the CSS's altitude. Δz_{int} is the centroid difference of the dual atom interference loops, which is the same as introduced in the centrifugal force effect. This effect induces a differential phase error of $(0.0 \pm 1.5) \times 10^{-4}$ rad.

The CSSAI uses sidebands generated by phase modulation as the Raman laser pairs. This can induce a phase shift due to the multiple-sideband effect (60). This phase is proportional to the value of gravity. Under microgravity condition in the CSS, this phase is suppressed by at least three orders compared to the case on the ground. Under actual experiment parameters, the induced phase for ^{85}Rb and ^{87}Rb are $\pm 2.0 \times 10^{-5}$ rad and $\pm 1.0 \times 10^{-5}$ rad. The induced differential phase is $(0.0 \pm 2.2) \times 10^{-5}$ rad.

The wavefront aberration of the Raman laser will cause distortion of the shearing fringe, and induce phase shift. Because the cold atom clouds overlap and have similar expansion rates, the induced phases are common for some extent for the dual-species interference fringes (61). The reflecting mirror of the Raman laser has an aberration better than $\lambda/10$. To estimate the induced differential phase, we numerically generate 10 random wavefronts with $\lambda/10$ peak-to-valley distortion for the mirror and calculate the induced differential phase. The result is $(0.0 \pm 1.2) \times 10^{-2}$ rad.

The uncertainty of the Raman laser's frequency and the interference time sequence will induce phase errors, which are proportional to the value of gravity. These phase errors are suppressed greatly due to the microgravity condition in the CSS. The laser frequency has an accuracy better than 1 MHz. The interference time sequence is referenced to a temperature-controlled crystal oscillator (TCXO), which has a frequency accuracy better than 1 part per million. The induced differential phase errors from these two effects are less than 1 μrad and can therefore be neglected.

Quantum Science and Technology-National Science and Technology Major Project grant 2021ZD0300604, Hubei Provincial Science and Technology Major Project grant ZDZX2022000001, Defense Industrial Technology Development Program grant JCKY2022130C012, National Natural Science Foundation of China grant 12174403, National Natural Science Foundation of China grant W2412045, National Natural Science Foundation of China grant U25D8014, Natural Science Foundation of Hubei Province grant 2022CFA096, Strategic Priority Research Program of the Chinese Academy of Sciences grant XDA0520501, and Strategic Priority Research Program of the Chinese Academy of Sciences grant XDB1070103.

Author contributions:

Conceptualization: X.C., J.W., and M.-S.Z. Methodology: X.C., D.F.Z., and J.-T.L. Software: X.C., D.-F.Z., and J.-T.L. Validation: X.L., J.-Q.Z., L.Q., H.-Y.S., D.-F.G., B.T., L.Z., R.-B.L., and Y.-B.W. Formal analysis: X.C., D.-F.Z., and J.-T.L. Investigation: X.C., D.-F.Z., J.-T.L., W.-Z.W., W.-H.X., and J.-Y.W. Resources: M.-Z.A., L.Q., X.-X.G., Q.-F.C., Y.T., Z.-F.L., X.-H.Y., and S.-Q.W. Data curation: D.-F.Z, W.-Z.W., and W.-H.X. Writing–original draft: X.C. and D.-F.Z. Writing–review and editing: X.C., J.W., and M.-S.Z. Visualization: X.C. and D.-F.Z. Supervision: X.C., J.W., and M.-S.Z. Project administration: M.-S.Z., J.W., and H.-E.Z. Funding acquisition: X.C., J.-Q.Z., L.Z., J.W., and M.-S.Z.

Competing interests:

The authors declare that they have no competing interests.

Data, code, and materials availability:

All data and code needed to evaluate and reproduce the results in the paper are present in the paper and/or the Supplementary Materials. This study did not generate new materials.

Supplementary Materials

This PDF file includes:

- Supplementary Text
- Figs. S1 to S6
- Table S1
- References

DOWNLOAD

2.21 MB

REFERENCES

1

M. S. Safronova, D. Budker, D. DeMille, D. F. J. Kimball, A. Derevianko, C. W. Clark, Search for new physics with atoms and molecules. *Rev. Mod. Phys.* **90**, 025008 (2018).

↶ GO TO REFERENCE

CROSSREF

WEB OF SCIENCE

GOOGLE SCHOLAR

2

S. Schlamminger, K. Y. Choi, T. A. Wagner, J. H. Gundlach, E. G. Adelberger, Test of the equivalence principle using a rotating torsion balance. *Phys. Rev. Lett.* **100**, 041101 (2008).

↶ GO TO REFERENCE

CROSSREF

PUBMED

WEB OF SCIENCE

GOOGLE SCHOLAR

3

J. G. Williams, S. G. Turyshev, D. H. Boggs, Lunar laser ranging tests of the equivalence principle. *Class. Quantum Grav.* **29**, 184004 (2012).

↶ GO TO REFERENCE

CROSSREF

WEB OF SCIENCE

GOOGLE SCHOLAR

SHOW ALL REFERENCES

eLetters (0)

eLetters is a forum for ongoing peer review. eLetters are not edited, proofread, or indexed, but they are screened. eLetters should provide substantive and scholarly commentary on the article. Neither embedded figures nor equations with special characters can be submitted, and we discourage the use of figures and equations within eLetters in general. If a figure or equation is essential, please include within the text of the eLetter a link to the figure, equation, or full text with special characters at a public repository with versioning, such as Zenodo. Please read our [Terms of Service](#) before submitting an eLetter.

LOG IN TO SUBMIT A RESPONSE

No eLetters have been published for this article yet.

PDF

CURRENT ISSUE



PDF

A Li metal-SiO_x hybrid anode enabling synergistic plating/alloying dual-mechanism lithium storage

BY CHAO-HUI ZHANG, JUAN ZHANG, *ET AL.*

Impaired liver-muscle lactate metabolism causes sarcopenia via lactic acidosis in skeletal muscle in mice

BY TAKAHIRO EGUCHI, NOZOMI HAYASHIJI, *ET AL.*

Amplification of RNA for identification of Zika and HCV in whole blood

BY JONGWON LEE, HANKEUN LEE, *ET AL.*

Sign up for ScienceAdviser

Get *Science*’s award-winning newsletter with the latest news, commentary, and research, free to your inbox daily.

SUBSCRIBE >

LATEST NEWS

SCIENCEINSIDER | 5 SEP 2026

Outrage greets NIH pact that could funnel biodefense research funds to Pentagon

SCIENCEINSIDER | 4 SEP 2026

Massive herbarium merger rescues century-old plant collection

SCIENCEINSIDER | 4 SEP 2026

Bernie Sanders aims to ban AI ‘superintelligence.’ But experts can’t agree on what the term means

NEWS | 4 SEP 2026

Are these killer whales out for revenge on humans—or just trying to play?

SCIENCEINSIDER | 4 SEP 2026

To prevent forest fires, Trump wants to end a road-building ban. Scientists are skeptical

SCIENCEINSIDER | 4 SEP 2026

NSF initiative seeks tech fixes for Colorado River water crisis

ADVERTISEMENT

RELATED JOBS

Ignite Scholar, Research Assistant Professor

Northeast Ohio Medical University
Rootstown, Ohio

Nuclear Science Division Director

Lawrence Berkeley National Laboratory
Berkeley, California

Faculty Positions (Information & Mathematical Science) in ShanghaiTech University

ShanghaiTech University
Shanghai (CN)

MORE JOBS ►

RECOMMENDED

RESEARCH ARTICLE | DECEMBER 2018

Interleaved atom interferometry for high-sensitivity inertial measurements

REPORT | NOVEMBER 2011

Twin Matter Waves for Interferometry Beyond the Classical Limit

RESEARCH ARTICLE | NOVEMBER 2019

Electrically driven optical interferometry with spins in silicon carbide

[View full text](#) | [Download PDF](#)

Science

Science
Advances

Science
Immunology

Science
Robotics

Science
Signaling

PDF

FOLLOW US



GET OUR NEWSLETTER

NEWS

- [All News](#)
- [ScienceInsider](#)
- [News Features](#)
- [Subscribe to News from Science](#)
- [News from Science FAQ](#)
- [About News from Science](#)
- [Donate to News](#)

COMMENTARY

- [Opinion](#)
- [Analysis](#)
- [Blogs](#)

AUTHORS & REVIEWERS

- [Information for Authors](#)
- [Information for Reviewers](#)

ADVERTISERS

CAREERS

- [Careers Articles](#)
- [Find Jobs](#)
- [Employer Hubs](#)

JOURNALS

- [Science](#)
- [Science Advances](#)
- [Science Immunology](#)
- [Science Robotics](#)
- [Science Signaling](#)
- [Science Translational Medicine](#)
- [Science Partner Journals](#)

LIBRARIANS

- [Manage Your Institutional Subscription](#)
- [Library Admin Portal](#)
- [Request a Quote](#)
- [Librarian FAQs](#)

RELATED SITES

ABOUT US

Leadership

Work at AAAS

Prizes and Awards

HELP

FAQs

Access and Subscriptions

Order a Single Issue

Reprints and Permissions

TOC Alerts and RSS Feeds

Contact Us

PDF

© 2026 American Association for the Advancement of Science. All rights reserved. AAAS is a partner of HINARI, AGORA, OARE, CHORUS, CLOCKSS, CrossRef and COUNTER. *Science Advances*
eISSN 2375-2548.

[Terms of Service](#) | [Privacy Policy](#) | [Cookie Preferences](#) | [Accessibility](#)

This website utilizes technologies such as cookies to enable essential site functionality, as well as for analytics, personalization, and targeted advertising. To learn more, view the following link: [Privacy Policy](#).